Week 3 Notes

Answers to the follow questions:

1. What is a graphical user interface (GUI)?
    - A graphical user interface is a set of programs that allow users to interact with the computer system, via icons, windows, and various other visual elements.
2. What is a desktop environment?
    - A desktop environment is an implementation of the desktop metaphor made of a bundle of programs running on top of a computer operating system, sometimes described as a graphical shell.
3. What is the command line interface (CLI)?
    - The user interacts with the computer by inputting commands, the commands are then interpreted by a shell which can be considered a programming language.
4. How do I access the command line interface (CLI)?
    - There are two ways to access the command line interface.
        - Terminal emulator
        - Linux console
5. What is a virtual console?
    - A virtual console is a terminal session that runs in Linux system memory.
6. What is a terminal emulator?
    - A terminal emulator is a program that allows you to access the Linux CLI when using the GUI.
7. What is bash?
    - Bash shell is a program that provides interactive access to the linux system.
8. What is the shell prompt?
    - Shell prompt is the line waiting for the user to type a command. The line the user see's in the screen, and will appear whenever the shell is ready to accept input.

Definition, usage , and example of the following commands:

1. clear

Definition:

Clears the terminal window/screen

Usage

"Clear' + "option"

Example

clear -x prevents clear from attempting to clear the scrollback buffer

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2. echo

Definition:

Prints/displays text to the screen

Usage

'echo' + 'option' + 'string'

Example

echo -n "hello world" Doesn't output a trailing newline

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3. date

Definition:

Displays current date and time.

Usage

'date' + 'option'

Example

date -d displays the time described by string

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4. free

Definition

Displays the amount of free and used memory in the system

Usage

'free' + 'option'

Example

free -h --giga shows output field in human readable format and in gigabytes

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5. uname

Definition

Show basic information about your system such as: kernel,OS, architecture, etc...

Usage

'uname' + 'option'

Example

uname -r prints the kernel release version

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6. history

Definition

Display a history of every command you run so far.

Usage

'history' + 'option'

Example

history # displays the entire command line history.

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7. man

Definition

Shows/displays the manual page for a given command.

Usage

'man' + 'option'

Example

man -a intro displays all available intro manual pages contained within the manual.

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8. tldr

Definition

Similar to cheat but is available via APT and keeps a local cache.

Usage

'tldr' + "option

Example

tldr -L override the language.

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9. cheat

Definition

A collection of cheat sheets

Usage

'cheat' + 'command'

Example

cheat git cheat sheet for git commands

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10. hostname

Definition

Shows the system hostname (computer name)

Usage

'hostname' + 'option'

Example

hostname -d display the name of the DNS domain

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11. df

Definition

Display the amount of disk space available/used on entire file systems/partition.

Usage

'df' + 'option'

Example

df -h-a displays the disk space in all systems in human readable format.

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12. du

Definition

Displays the amount of space used by specific files or directories

Usage

'du' + 'option'

Example

du -c produce a grand total of space used

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13. figlet

Definition

Displays text in large ASCII letters

Usage

'figlet' + 'option'

Example

figlet -f select the font